

MTH208

Examination - January Semester 2026

Advanced Linear Algebra

Tuesday, 21 April 2026

04:00 pm - 06:00 pm

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **FIVE (5)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded if **FULL** working is shown.
4. This is an **Open-book** examination.

At the end of the examination.

Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.

If you have used more than one answer book, please tie them together with the string provided.

**THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR
SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.**

(Full marks: 100)

Question 1

You are given the following set of vectors in $\mathbb{R}^3$:

$$\mathfrak{C} = \left\{ \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right\}.$$

A linear operator L acts on $\mathbb{R}^3$ as shown below.

$$L(u) = Au, u \in \mathbb{R}^3$$

$$\text{where } A = \begin{pmatrix} 0 & 1 & 2 \\ -3 & 5 & 4 \\ 3 & 0 & -1 \end{pmatrix}.$$

Question 1a

Show that $\mathfrak{C}$ forms a basis in $\mathbb{R}^3$.

(4 marks)

Question 1b

What is the change-of-basis matrix from the standard basis $\mathfrak{B}$ to the basis $\mathfrak{C}$?

(4 marks)

Question 1c

Compute the matrix representation of L on the basis $\mathfrak{C}$.

(7 marks)

Question 2

The matrix below has eigenvalues 9 and $6 - 3i$.

$$B = \begin{pmatrix} 7 - 2i & 3 & 2 + 2i \\ 0 & 9 & 0 \\ 1 + i & 3 & 8 - i \end{pmatrix}.$$

Question 2a

Calculate the Jordan canonical form of B .

(9 marks)

Question 2b

Find the eigenvectors, and generalised eigenvectors if any, belonging to the eigenvalue 9 of B . Leave your answer in the most general form.

(11 marks)

Question 3a

An inner product is defined on the real vector space $\mathbb{R}^2$ as

$$[u, v] = v^T G u \quad ; \quad u, v \in \mathbb{R}^2,$$

where G is a positive definite matrix given by

$$G = \begin{pmatrix} a & b \\ b & a \end{pmatrix} \quad ; \quad a, b \in \mathbb{R}.$$

Question 3a(i)

Determine the eigenvalues and *normalised* eigenvectors of G .

(5 marks)

Question 3a(ii)

Employ a property of positive definite matrices to show that G is positive definite if and only if $a > |b|$.

(2 marks)

Question 3a(iii)

For $a = 5, b = -4$, find the positive-definite square-root of G .

Hence, find a matrix K such that for all $u, v \in \mathbb{R}^2$,

$$[u, v] = \langle u', v' \rangle$$

where

$$u' = Ku$$

$$v' = Kv$$

and $\langle \cdot, \cdot \rangle$ denotes the standard inner product on $\mathbb{R}^2$.

(8 marks)

Question 3b

Compute the singular value decomposition (SVD) of

$$S = \begin{pmatrix} 5 & 0 \\ -2 & 6 \\ -5 & 0 \end{pmatrix}.$$

(10 marks)

Question 4

The matrix T below is a linear operator that acts in the complex vector space $\mathbb{C}^3$ with the standard inner product.

$$T = \begin{pmatrix} \frac{i}{2} & \frac{3i}{2} & \frac{i}{\sqrt{2}} \\ \frac{3i}{2} & \frac{i}{2} & \frac{-i}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{-i}{\sqrt{2}} & -i \end{pmatrix}.$$

The left polar decomposition of T is shown below.

$$T = UP$$

where U is a unitary matrix and P is a semipositive definite matrix.

Question 4a

Demonstrate that T is a normal operator, and explain briefly why T is diagonalisable.

(5 marks)

Question 4b

Calculate the kernel and range of T explicitly and show that they are orthogonal complements.

(6 marks)

Question 4c

T has three eigenvalues, two of which are $2i$ and $-2i$ with eigenvectors $(1 \ 1 \ 0)^T$ and $(-1 \ 1 \ \sqrt{2})^T$ respectively. From your answer in Question 4b, what is the third eigenvalue and an eigenvector corresponding to it?

Hence, compute P and explain why U is not uniquely determined by the polar decomposition.

(14 marks)

Question 5a

Let

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 2 \end{pmatrix} \text{ and } v_2 = \begin{pmatrix} 2 \\ 3 \\ 1 \\ 1 \end{pmatrix}.$$

Denote the subspace of $\mathbb{R}^4$ spanned by v_1 and v_2 by W . Compute a basis of linear functionals in the annihilator of W . Viewing the action of linear functionals on $\mathbb{R}^4$ as usual matrix multiplication of 1×4 matrices by 4×1 matrices, express your final answers in terms of row vectors.

(7 marks)

Question 5b

Let $\mathcal{B} = \{w_1, w_2, w_3\}$ be the basis for $\mathbb{R}^3$ defined by

$$w_1 = (1, 0, -1), w_2 = (1, 1, 1), w_3 = (-1, -1, 0).$$

Find the dual basis of $\mathcal{B}$.

(8 marks)

----- END OF PAPER -----